RISHI THAKUR

ADVANCED DATA SCIENCE ASSOCIATE

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PROFESSIONAL SUMMARY

Data Scientist with a quantitative background (M.Tech CSE from IIIT Naya Raipur, GATE CS qualified) and expertise in statistical inference, machine learning, and predictive modeling. Experienced deploying generative diffusion models, neural networks, and automated SQL pipelines. Skilled at translating complex analytical findings into executive dashboards for strategic decision-making.

SKILLS

- **Machine Learning & Statistics:** Statistical Inference, Hypothesis Testing, Predictive Modeling, High-Dimensional Data, Resampling (SMOTE + Tomek Links), Feature Engineering, Random Forest, XGBoost, Regression
- **Generative AI & Deep Learning:** Generative Diffusion Models (DDPM/DDIM), Deep Learning (PyTorch), Recurrent Encoders (GRU), Transformers, Text Mining (BERT Embeddings), Scikit-learn
- **Data Systems & Analytics:** Python (Pandas, NumPy), Advanced SQL (Joins, CTEs, Window Functions), MySQL, MariaDB, OracleDB, Microsoft Power BI (DAX), Linux/Bash, Git
- **Communication & Spoken Skills:** Client Presentations, Statistical Interpretation for Stakeholders, Executive Briefings, Cross-Functional Collaboration, Technical Writing

WORK EXPERIENCE

Chhattisgarh Infotech Promotion Society (CHiPS) | Raipur, India | 06/2025 – 03/2026
Data Science & Analytics Intern • Dept. of Health & Family Welfare, Govt. Of Chhattisgarh

- Conducted exploratory data analysis (EDA) and hypothesis testing across **33 districts** on healthcare telemetry, resolving schema anomalies.
- Engineered an automated **SQL/ETL feature pipeline** mapping AEBAS biometric IDs to State Treasury payroll codes with **100% data fidelity**.
- Formulated an analytical **3-tier risk classification model** tracking bonded medical personnel across facilities, preventing public fund leakage.
- Reconciled disparate backend database systems across MariaDB, State Treasury, and modular Next-Gen e-Hospital schemas, eliminating data silos.
- Architected an executive 6-page **Power BI dashboard** structuring **4-tier drill-down hierarchies** (District → Facility → Designation → Individual) with dynamic DAX.

KEY PROJECTS

Proactive 5G Handover Prediction (Diffusion Models) • *M.Tech Thesis — Predictive Modeling & Telemetry*
Technologies: Python, PyTorch, Diffusion Models (DDPM/DDIM), GRU, Scikit-learn, Random Forest, SMOTE, Tomek Links

- Developed a predictive engine pairing a **GRU encoder** with a **generative Diffusion Model (DDPM/DDIM)** simulating signal trajectories from multivariate **spatio-temporal** telemetry.
- Addressed extreme **98%/2% class imbalance** using a hybrid **SMOTE + Tomek Links** pipeline, training a Random Forest policy achieving a **97.64% F1-score**.

AI vs. Human Text Classification Pipeline • *Applied NLP & Deep Learning*
Technologies: Python, BERT, XGBoost, Bi-LSTM, Transformers, 1D-CNN, Scikit-learn

- Engineered dual text mining architectures: extracted 768-dim dense **BERT embeddings** paired with **XGBoost**, and constructed a deep hybrid Bi-LSTM/Transformer network.
- Applied stratified sampling (80/20), text normalization (NFKD), dropout (0.5), and learning rate decay, achieving **99% Precision** and **98% Recall** in synthetic LLM classification.

EDUCATION

M.Tech in Computer Science & Engineering | IIIT Naya Raipur (CGPA: 7.90) | 08/2024 – 07/2026
B.Tech in Computer Science | Amity University Chhattisgarh (CGPA: 7.71) | 08/2019 – 06/2023

CERTIFICATIONS & KEY QUALIFICATIONS

- **Qualified GATE Computer Science (CS)**, 2024
- **Python for Data Science & AI** (Coursera)
- **Celonis Process Mining Foundations** (Credly)
- **Microsoft Power BI** (Udemy)
- **SQL & Database Design** (Udemy)
- **IELTS Academic:** 7.5 Band Score